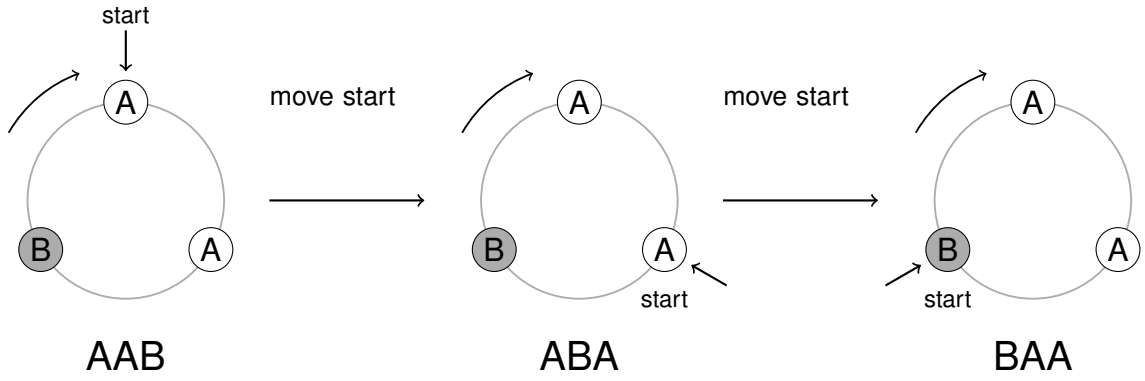
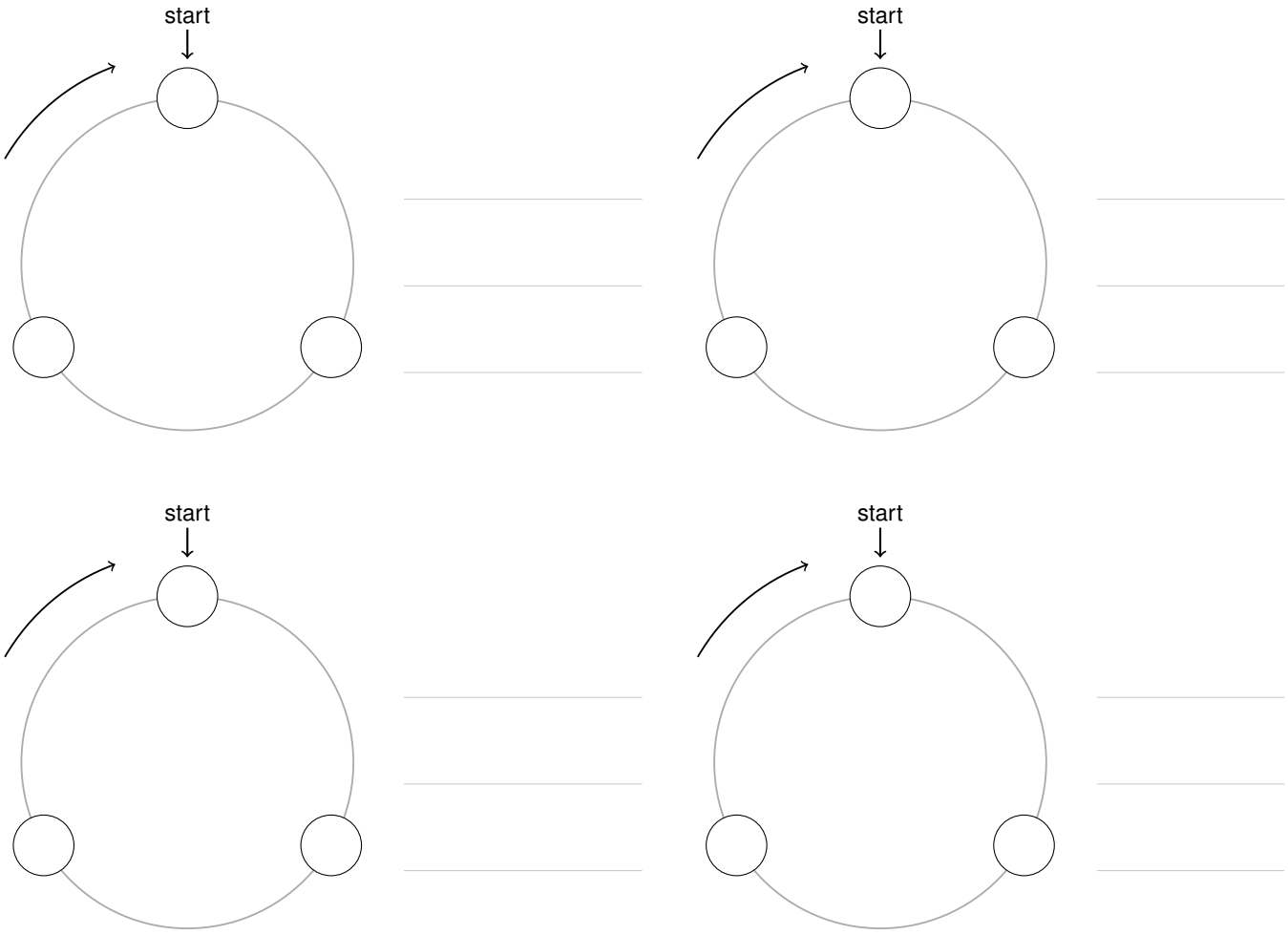
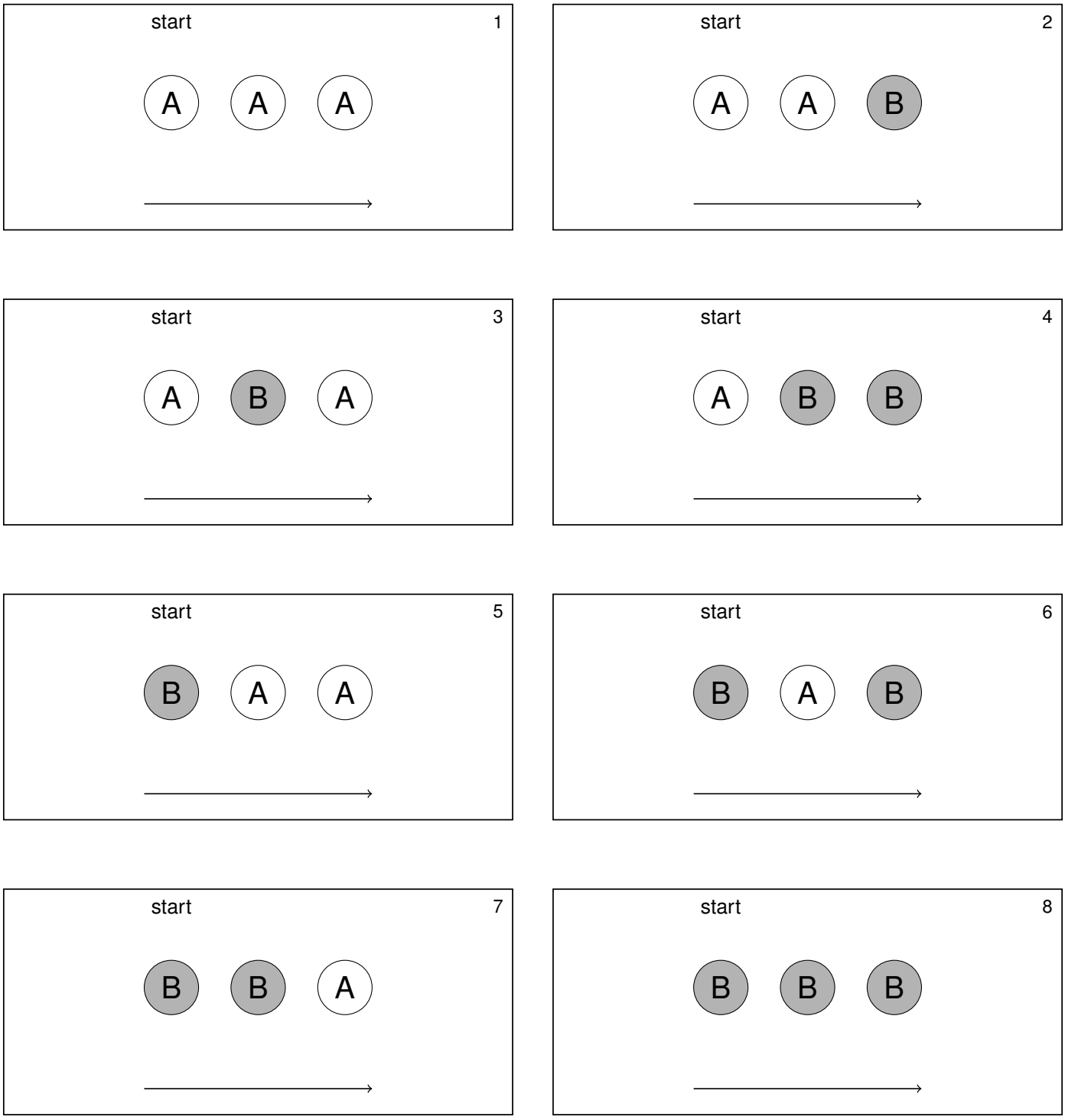
Each bead may be A or B; one-color rings are allowed. Read clockwise from the start arrow. The bead order is the readout. Rings count as the same when turning makes them match. Keep them face up; do not flip them over.



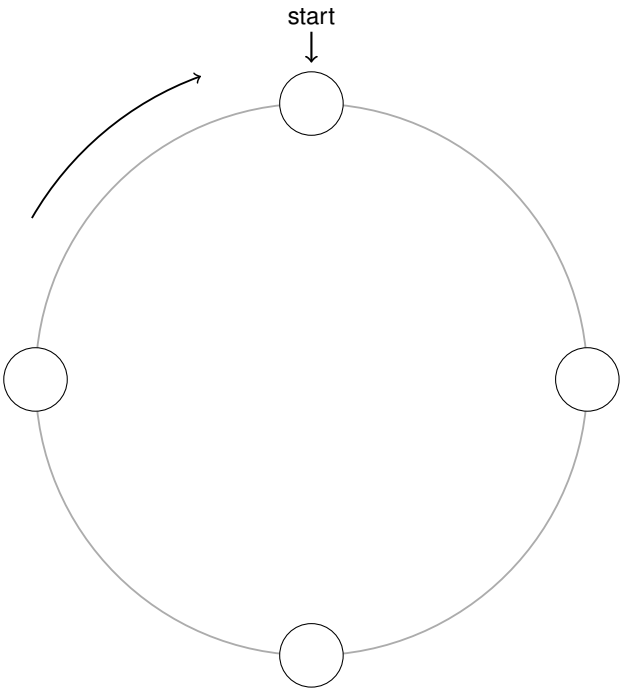
Problem 1: Find every three-bead ring using A and B. How many different readouts does each ring have?

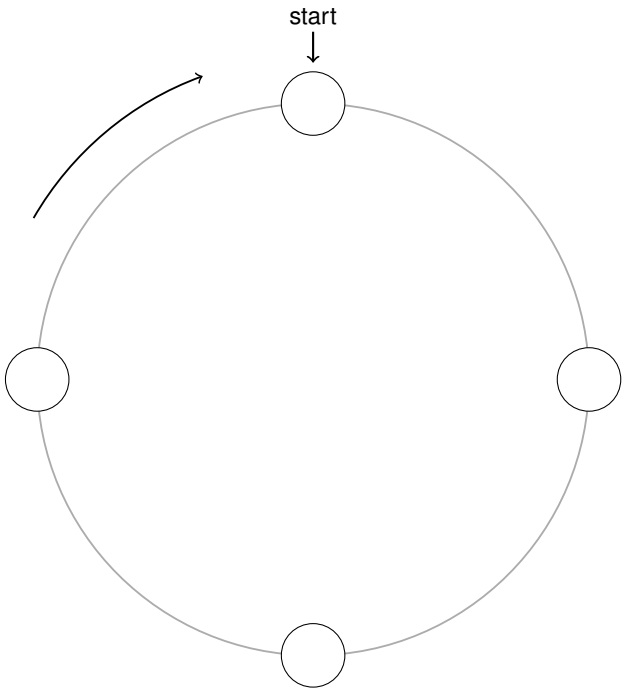


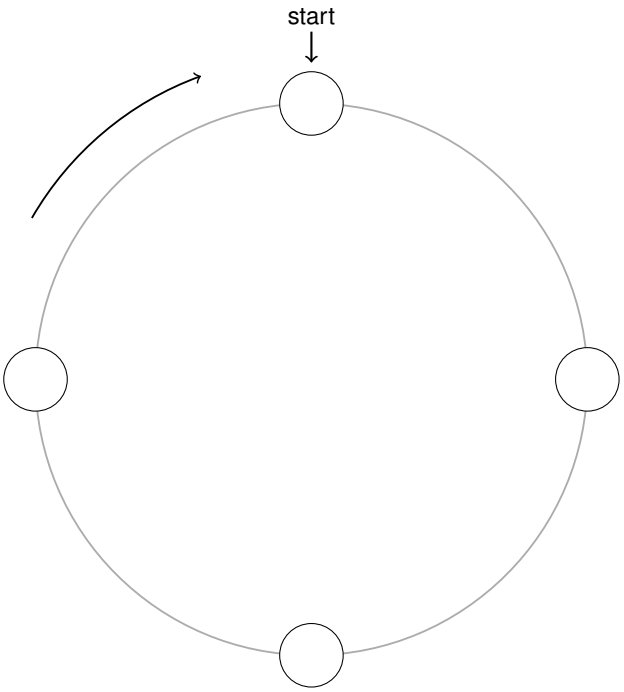
Problem 2: Group the eight cards by turning. Someone divides 8 by 3 to count the rings. What went wrong?

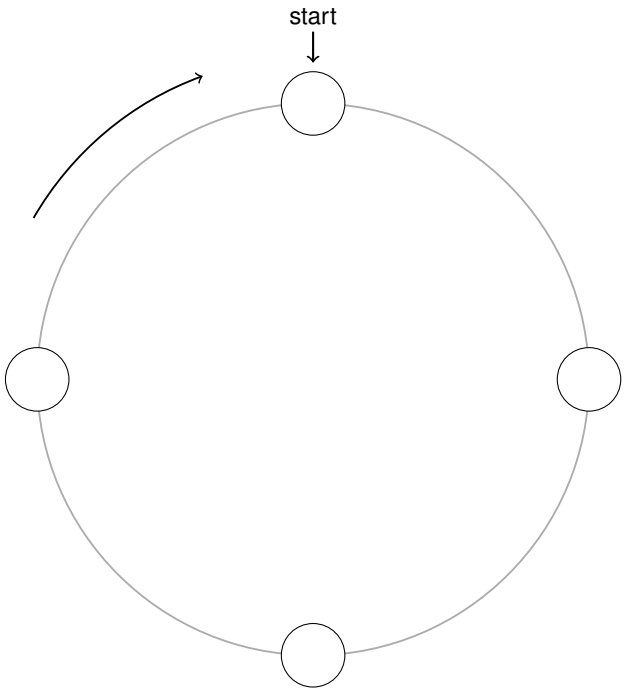


Problem 3: Find four-bead rings with one, two, three or four different readouts, or explain why a requested number is impossible. What feature lets a mixed ring repeat early?

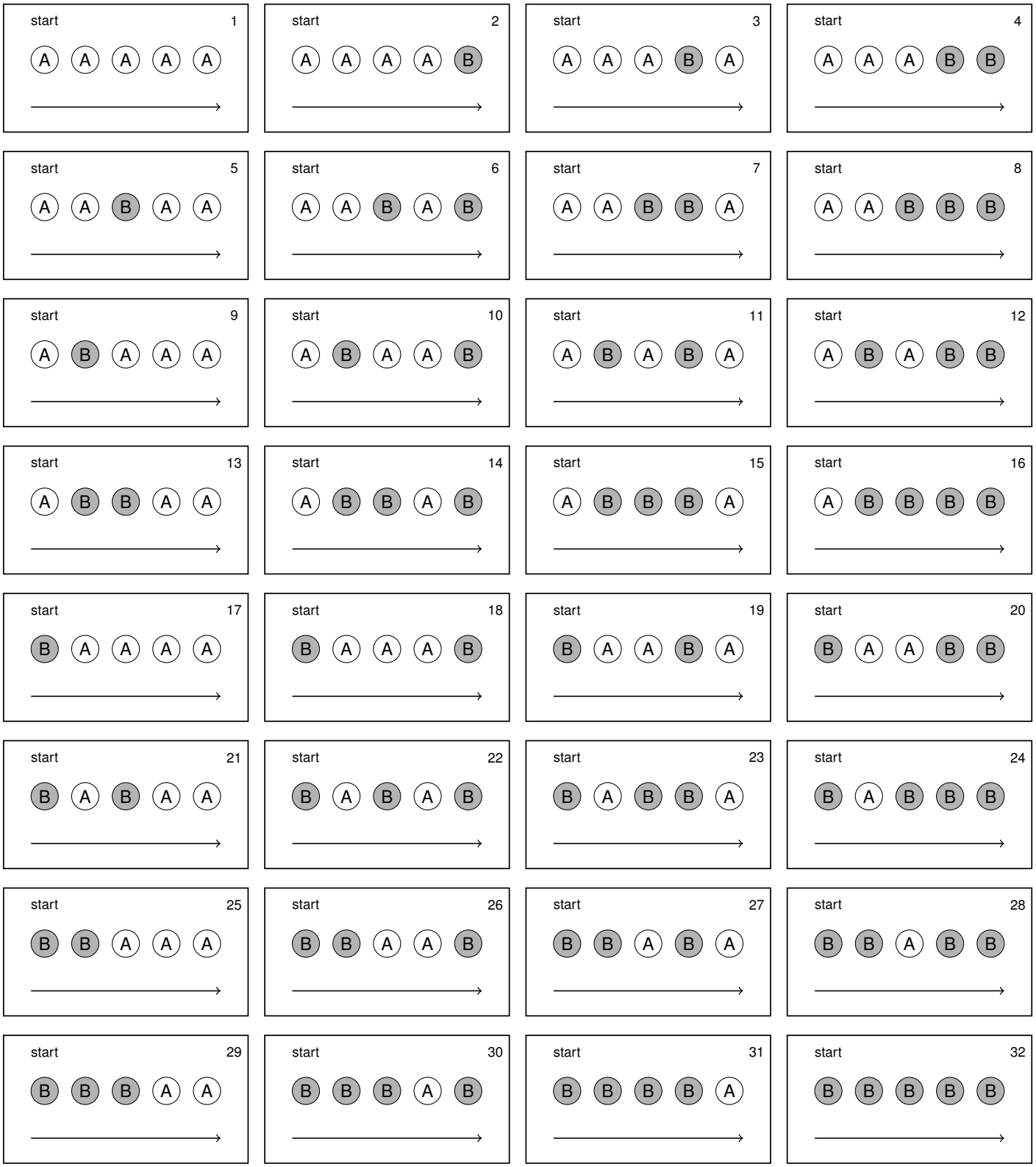




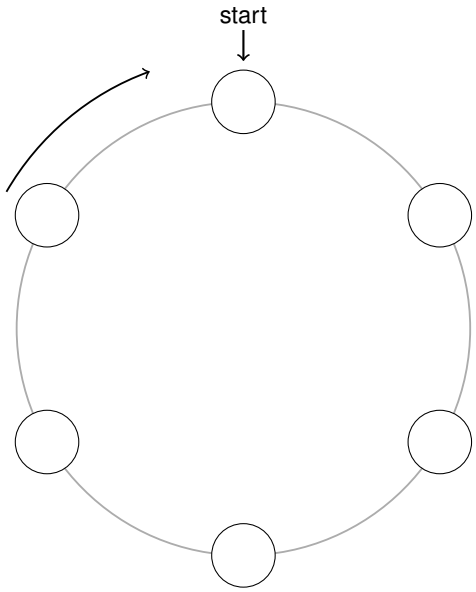




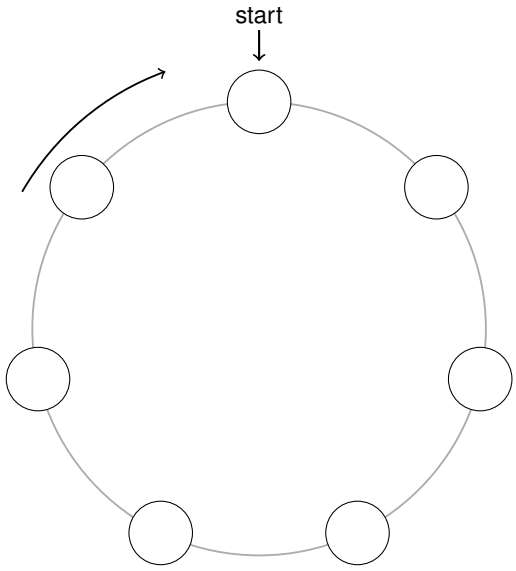
Problem 4: Group the 32 cards by turning. Count the rings with a calculation, and explain why the group sizes differ from the four-bead case.



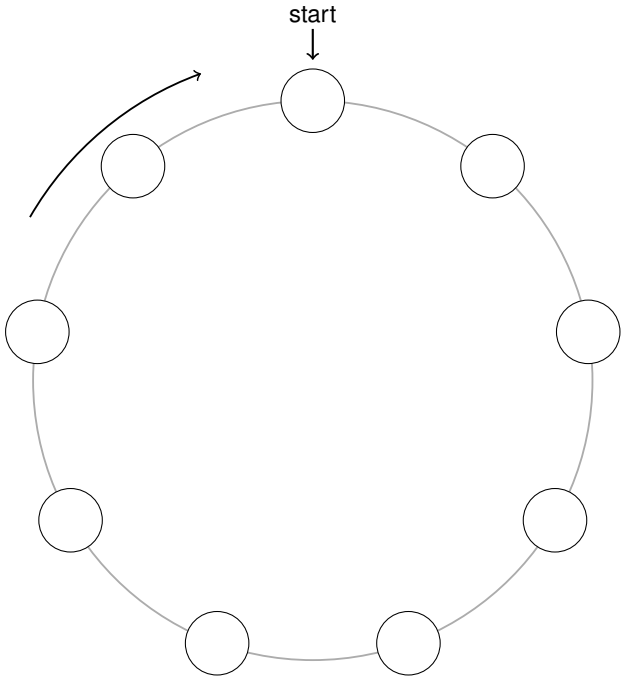
Problem 5: Which numbers of different readouts can a ring using both A and B have on six beads? On seven? On nine?



6 beads



7 beads



9 beads

Problem 6: A prime number has exactly two positive divisors: 1 and itself. What numbers of different readouts are possible for a ring of prime length that uses at least two colors? Find a rule and explain it.

Problem 7: How many five-bead rings can be made when each bead may be any of three colors? Find a counting rule for c colors and any prime length p, and explain what it tells you about divisibility.